

NEC Math Companion Syllabus

Mathematical Foundations for the NEC / MoE Thesis

Companion to NEC_thesis_syllabus.md

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Purpose	Deepen only the math that supports the NEC thesis
Orientation	Connect new ML math to prior physics/math background
Main split	Proposal-level math first; thesis-depth math after proposal

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1 Verdict and Scope

You should go deeper in math, but not equally everywhere. The NEC thesis needs a clear mathematical spine:

1. Mixture-model probability.
2. Optimization and training dynamics.
3. Load balancing and expert collapse.
4. HMM / Markov-switching math.
5. Financial validation statistics.

Information theory and ensemble bias-variance are useful supporting material. They should appear if they help explain gates, entropy, regularization, and why multiple experts might help. They should not become the main thesis.

Before proposal: understand and implement the math. After proposal: deepen derivations for the thesis and defense.

2 Connection to the Current Syllabus

Current syllabus module	Math topic	Thesis use
Module 2: Linear algebra	Matrix calculus, projections, SVD	Regression, PCA, gradients, softmax gate
Module 3: Probability/statistics	Conditional probability, likelihood, entropy	MoE probability, losses, gate diagnostics
Module 4: Linear/logistic models	Softmax, convex losses, regularization	NEC classifier/gate math
Module 5: Validation	Sampling error, purging, multiple testing	Defend empirical claims
Module 6: Neural networks	Backprop, optimization	Train corrected NEC
Module 7: Sequence models	Recurrences, hidden states	LSTM gate encoder
Module 8: Mixture models	Latent variables, EM	Mathematical heart of MoE
Module 9: HMMs/regimes	Dynamic programming, Markov chains	Hamilton baseline
Module 10: MoE theory	Routing, load balancing	Corrected NEC design
Module 11: Empirical asset pricing	Factor math, residual returns	Finance interpretation

3 Bridge From Previous Math

You are not starting from zero. Several pieces of your physics/math background transfer directly:

- Linear algebra / tensors -> neural network layers, PCA, covariance matrices, embeddings, matrix calculus.
- Calculus and variational principles -> optimization, regularized objectives, Lagrangian penalties, gradient descent.
- ODEs / dynamical systems -> state evolution intuition, though HMMs are stochastic state models rather than deterministic systems.
- Sturm-Liouville / spectral thinking -> useful for covariance and PCA intuition, but not the center of NEC.
- Green's functions / propagators -> useful analogy for forward recursions, but HMM forward-backward is discrete probabilistic recursion.
- Quantum mechanics / statistical mechanics -> useful analogy for softmax, partition functions, entropy, and variational free energy.
- Probability -> direct foundation for latent-variable models, mixture likelihoods, and regime inference.

The new math is less physics-flavored and more statistical ML-flavored: conditional likelihoods, latent variables, gates, EM, regularization, and financial validation.

4 Depth Map

Topic	Pre-proposal depth	Thesis-depth after proposal	Priority
Mixture-model probability	Derive MoE equation and implement soft MoE	Likelihood derivation, EM comparison, log-sum-exp, responsibilities	Highest
Optimization	Backprop, Adam, regularization, gradient flow through gates	Nonconvex training, collapse dynamics, penalty methods	Highest
Load balancing	Know formula and implement it	Derive auxiliary loss, analyze utilization/entropy tradeoff	Highest
HMM / dynamic programming	Implement forward algorithm and understand Hamilton filter	Forward-backward, Viterbi, Baum-Welch, Markov-switching regression	High
Financial validation statistics	Implement purging, rank-IC, backtest costs	Multiple testing, deflated Sharpe, uncertainty of IC	High
Information theory	Entropy, KL, cross-entropy	Gate entropy, mutual information diagnostics if useful	Medium

Bias-variance / ensembles	ESL-level bias-variance	Optional expert-ensemble discussion	Low-medium
Stochastic calculus	Skip	Skip unless supervisor requires it	Low
Measure theory	Skip	Skip	Low

5 Pillar 1: Linear Algebra and Matrix Calculus

Connected current modules: Module 2, Module 4, Module 6.

Why it matters. MLP experts, LSTM gates, softmax gates, ridge/logistic baselines, PCA, and covariance diagnostics are all linear algebra plus nonlinearities.

Previous-knowledge bridge. Inner products, basis changes, eigenvectors, and quadratic forms already show up in physics. Here the same objects become regression, covariance, and gradient machinery.

Readings by section number.

- Strang, *Linear Algebra and Its Applications*: Sec 3.2-3.4, 5.1-5.3, 6.1-6.3.
- Matrix Cookbook: Sec 2.1, 2.4, 2.5, 8.1, 8.2.
- ESL: Ch 3.2-3.4.

Focused exercise set.

- Strang Sec 3.3: Exercises 1, 3, and 7.
- Strang Sec 6.3: Exercises 1 and 3.
- ESL Ch 3: Exercises 3.4 and 3.6.
- Applied: derive and implement ridge regression from scratch.
- Applied: derive the softmax Jacobian and verify it numerically.

Thesis deliverable. You should be able to write

$$\pi(x) = \text{softmax}(Wh(x) + b)$$

and explain how gradients flow from the prediction loss through $\pi(x)$ into the gate parameters.

6 Pillar 2: Mixture-Model Probability

Connected current modules: Module 3, Module 8, Module 10.

Why it matters. This is the mathematical heart of the thesis. A proper MoE is a conditional mixture model:

$$p(y | x) = \sum_k p(z = k | x)p(y | x, z = k).$$

Previous-knowledge bridge. This extends ordinary probability and Bayes rule. The physics analogy is summing over latent states: the final prediction marginalizes over hidden expert assignments.

Readings by section number.

- Bishop Ch 2.1-2.3.
- Bishop Ch 9.1-9.4.
- ESL Ch 8.5.

Focused exercise set.

- Bishop Ch 2: Exercise 2.20.
- Bishop Ch 9: Exercises 9.7, 9.12, and 9.15.
- ESL Ch 8: Exercise 8.1 or 8.2.
- Applied: implement a 2-component GMM and compare with sklearn.
- Applied: implement a 3-expert soft MoE on synthetic regime data.

Thesis deliverable. You should be able to derive

$$\mathcal{L}(\theta) = - \sum_i \log \sum_k \pi_k(x_i; \theta) p(y_i \mid x_i, z_i = k; \theta)$$

and explain EM, gradient descent, hard routing, and soft routing.

7 Pillar 3: Information Theory for Gates

Connected current modules: Module 3, Module 10.

Why it matters. You need enough information theory to talk about gate entropy, cross-entropy loss, KL divergence, expert collapse, and whether the gate uses all experts.

Previous-knowledge bridge. Entropy connects naturally to statistical mechanics. Softmax can be read as a Boltzmann-like distribution over experts.

Readings by section number.

- Bishop Ch 1.6.
- Goodfellow Ch 3.13.
- Shazeer et al. (2017), Sec 4.

Focused exercise set.

- Bishop Ch 1: Exercises 1.10 and 1.11.
- Applied: compute entropy of MoE gate probabilities over a batch.
- Applied: plot gate entropy over training and identify collapse.
- Applied: add an entropy penalty and compare it with Shazeer load balancing.

Thesis deliverable. You should be able to report

$$H(\pi(x)) = - \sum_k \pi_k(x) \log \pi_k(x)$$

and interpret low entropy, high entropy, and healthy expert specialization.

8 Pillar 4: Optimization and Training Dynamics

Connected current modules: Module 4, Module 6, Module 10.

Why it matters. MoE success often depends less on architectural elegance and more on training dynamics. Gates can collapse; experts can fail to specialize; hard routing can block gradient flow.

Previous-knowledge bridge. A training objective is an energy functional; regularization and load balancing are penalty terms; optimization searches for parameters that reduce the objective.

Readings by section number.

- Prince Ch 6.1-6.5 and Ch 7.1-7.5.
- Goodfellow Ch 8.1-8.5.
- Goodfellow Ch 7.1-7.12.

Focused exercise set.

- Prince Ch 7: Problems 7.1 and 7.2.
- Goodfellow Ch 6: Exercises 6.1 and 6.2.
- Applied: derive backprop for a 2-layer MLP and verify with `torch.autograd`.
- Applied: train the same MoE with and without load balancing and compare expert utilization.

Thesis deliverable. Explain why hard argmax routing is problematic for gradient flow, why soft routing trains end-to-end, why MoE loss is nonconvex, and why auxiliary losses are penalty methods.

9 Pillar 5: Load Balancing and Expert Collapse

Connected current modules: Module 10, Thesis Research Design.

Why it matters. Without balancing, the gate can route most samples to one expert. Then the model becomes a single predictor with extra unused parts.

Previous-knowledge bridge. This is a constrained optimization idea: low prediction loss plus a non-degenerate allocation of samples across experts. The auxiliary loss acts like a Lagrange-style penalty.

Readings by section number.

- Shazeer et al. (2017), Sec 4.
- Fedus, Zoph, Shazeer (2022), Sec 2.2-2.3.
- Goodfellow Ch 7.1-7.2.

Focused exercise set.

- Paper exercise: reproduce Shazeer’s auxiliary load-balancing loss from Sec 4.
- Paper exercise: explain why expert utilization and average gate probability both matter.
- Applied: implement the load-balancing term.
- Applied: run no balancing, weak balancing, and strong balancing experiments.

Thesis deliverable. Include a table varying the balancing coefficient and reporting rank-IC, expert entropy, expert utilization, and turnover.

10 Pillar 6: Dynamic Programming, HMMs, and Hamilton

Connected current modules: Module 9, Module 11.

Why it matters. The HMM/Hamilton baseline is the classical finance comparison. You need enough dynamic programming to understand latent regime inference.

Previous-knowledge bridge. Think of this as a discrete-time state evolution problem. Unlike deterministic dynamical systems, the state is hidden and probabilistic. The forward algorithm propagates a probability vector through time.

Readings by section number.

- Bishop Ch 13.1-13.3.
- Tsay Ch 4.1-4.6.
- Hamilton (1989), Sec 1-4.

Focused exercise set.

- Bishop Ch 13: Exercises 13.1, 13.3, and 13.5.
- Tsay Ch 4: two end-of-chapter exercises tied to Markov switching or volatility regimes.
- Applied: implement the HMM forward algorithm.
- Applied: fit a two-state model to SPY returns and compare states with realized volatility.

Thesis deliverable. Explain Hamilton filtering $p(s_t \mid y_1, \dots, y_t)$ and contrast it with MoE gating $p(z = k \mid x_t)$. The key distinction is temporal Markov persistence versus feature-conditioned routing.

11 Pillar 7: Financial Validation Statistics

Connected current modules: Module 5, Module 11, Module 12.

Why it matters. A complicated MoE is not convincing if the validation leaks information.

Previous-knowledge bridge. This is applied statistics: sampling error, dependence, multiple testing, and estimating uncertainty under non-i.i.d. data.

Readings by section number.

- ESL Ch 7.1-7.12.
- Lopez de Prado AFML Ch 7.1-7.6.
- Bailey and Lopez de Prado (2014), Sec 1-4.

Focused exercise set.

- ESL Ch 7: Exercises 7.1, 7.2, and 7.10.
- Applied: implement purging and embargoing for a forward-return label.
- Applied: compute daily rank-IC and IC information ratio.
- Applied: simulate many random strategies and show why the best Sharpe is biased upward.

Thesis deliverable. Justify chronological splits, purging and embargoing, rank-IC rather than only MSE, transaction costs, and multiple-testing caution.

12 Optional Math: Bias-Variance for Experts and Ensembles

Verdict. Learn standard ESL bias-variance before the proposal. Save non-independent ensemble bias-variance for after the proposal, and only include it if it helps the thesis discussion.

Readings by section number.

- ESL Ch 2.9.
- ESL Ch 8.7 if you want ensemble context.

Focused exercise set.

- ESL Ch 2: Exercise 2.9 or one nearby bias-variance exercise from your edition.
- Applied: train several experts with different seeds and measure prediction correlation.
- Applied: compare an average ensemble to a gated MoE.

Thesis deliverable. Use this to support a short discussion: MoE is not just an ensemble; it is a conditional ensemble whose weights depend on x .

13 Before-Proposal Math Plan

13.1 Block A: NEC Probability and Gate Math

ESL Ch 2.1-2.9; Bishop Ch 2.1-2.3; Bishop Ch 9.1-9.4; ESL Ch 8.5. Implement GMM and soft MoE on synthetic data.

13.2 Block B: Optimization and Deep Learning

Prince Ch 6.1-6.5 and 7.1-7.5; Goodfellow Ch 8.1-8.5. Derive 2-layer backprop, implement a PyTorch MLP, and verify gradients.

13.3 Block C: HMM / Hamilton Baseline

Bishop Ch 13.1-13.3; Tsay Ch 4.6; Hamilton (1989), Sec 1-4. Implement the HMM forward algorithm and fit a two-state volatility model to SPY.

13.4 Block D: Financial Validation

ESL Ch 7.1-7.12; AFML Ch 7.1-7.6. Implement purged walk-forward splitting, rank-IC, and a decile backtest.

13.5 Proposal-Level Math Deliverables

Before proposal, be able to write and defend:

1. MoE conditional mixture equation.
2. Original NEC hard-routing defect.
3. Corrected soft-routing equation.

4. Load-balancing loss at a high level.
5. HMM/Hamilton baseline at a high level.
6. Walk-forward validation and purging logic.

14 After-Proposal Math Plan

This is where the deeper theory belongs.

Deepen mixture-model theory: responsibilities, EM versus gradient descent, Gaussian mixture likelihood, mixture density network connection, identifiability, and label switching.

Deepen optimization and load balancing: expert collapse dynamics, gate entropy, Shazeer versus Switch load balancing, penalty-method interpretation, and balancing coefficient sensitivity.

Deepen HMM and regime-switching: forward-backward smoothing, Viterbi path, Baum-Welch updates, Markov-switching regression, regime persistence, and transition matrix interpretation.

Deepen financial statistics: deflated Sharpe, multiple testing, confidence intervals for IC, robustness across market regimes, and transaction-cost sensitivity.

14.1 Thesis-Level Math Deliverables

By January 2027, the thesis should include:

1. A formal MoE model section.
2. A short EM/HMM comparison section.
3. A corrected NEC architecture section.
4. A load-balancing and gate-diagnostics section.
5. A validation/statistical testing section.

15 What Not To Overlearn

Do not spend major time on these unless your supervisor specifically pushes you there:

- stochastic calculus,
- measure-theoretic probability,
- PDE theory,
- advanced random matrix theory,
- optimal transport,
- PAC-Bayes,
- deep information theory.

16 Final Math Thesis Spine

The thesis math can be organized as:

1. Conditional mixture probability: $p(y | x) = \sum_k p(z = k | x)p(y | x, z = k)$.
2. NEC defect: three-class gate $\rightarrow$ binary hard route.
3. Corrected MoE: $\pi_k(x) = \text{softmax}(g_k(x))$, $\hat{y} = \sum_k \pi_k(x)f_k(x)$.
4. Training objective: prediction loss plus α times load-balancing loss.
5. Classical baseline: HMM / Hamilton filter with Markov state persistence.
6. Financial evaluation: walk-forward validation, purging, rank-IC, transaction-cost backtest.

If you can explain and implement those six pieces, the math foundation is strong enough for the proposal. The deeper derivations can be added during the thesis phase.